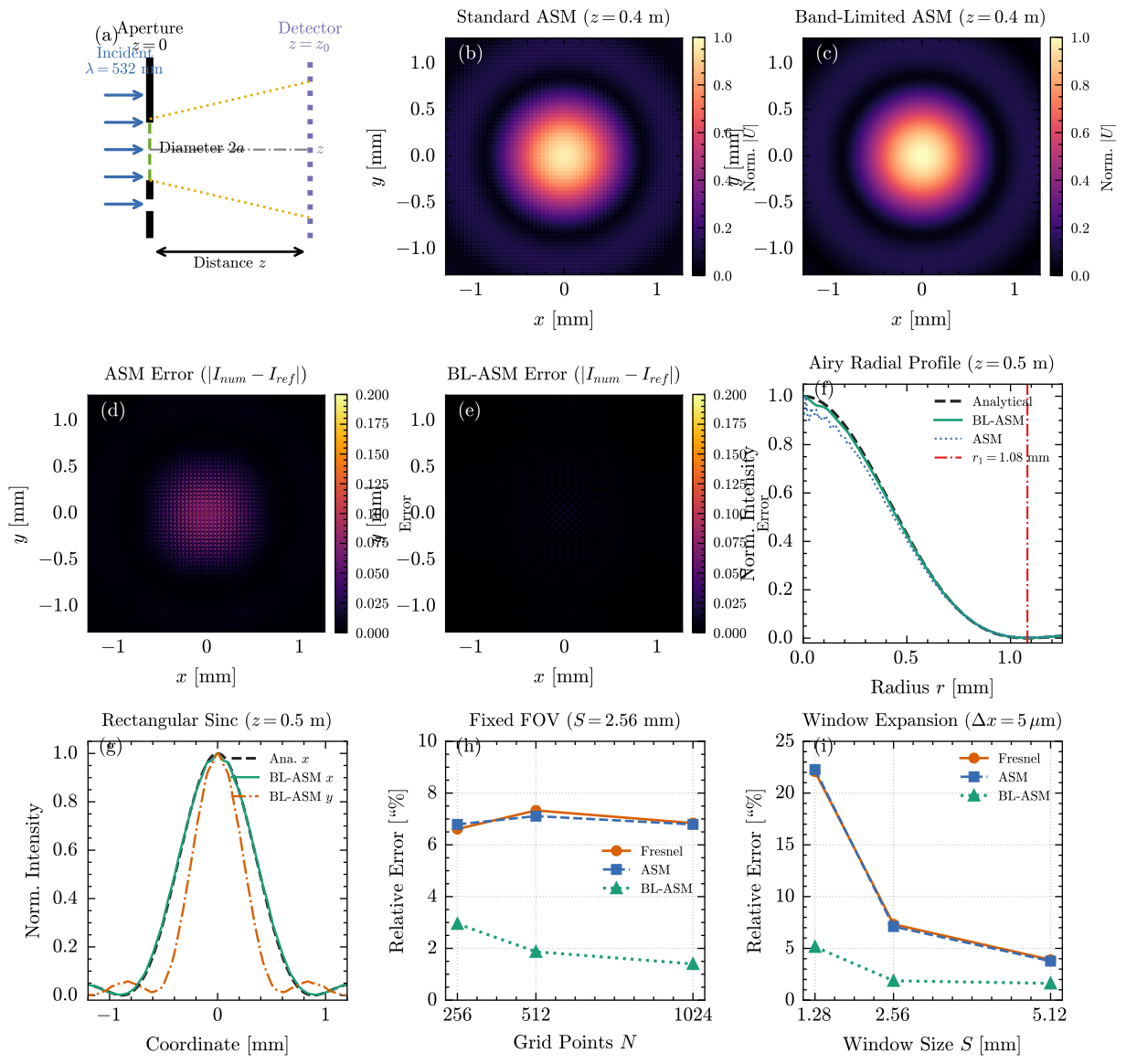

(a)
Aperture
$z = 0$
Incident
$\lambda = 532$ nm
Detector
$z = z_0$
Diameter $2a$
z
Distance z
Standard ASM ($z = 0.4$ m)
(b)
y [mm]
x [mm]
Norm. $|U|$
Band-Limited ASM ($z = 0.4$ m)
(c)
ASM Error ($|I_{num} - I_{ref}|$)
(d)
BL-ASM Error ($|I_{num} - I_{ref}|$)
(e)
Error
Airy Radial Profile ($z = 0.5$ m)
(f)
Analytical
BL-ASM
ASM
$r_1 = 1.08$ mm
Norm. Intensity
Radius r [mm]
Rectangular Sinc ($z = 0.5$ m)
(g)
Ana. x
BL-ASM x
BL-ASM y
Norm. Intensity
Coordinate [mm]
Fixed FOV ($S = 2.56$ mm)
(h)
Fresnel
ASM
BL-ASM
Relative Error [“%]
Grid Points N
Window Expansion ($\Delta x = 5\,\mu$m)
(i)
Fresnel
ASM
BL-ASM
Relative Error [“%]
Window Size S [mm]